GTC4Lusso USA - Electrical system absorption test - Latest Update: 2021-01-26

F2.13 Electrical system absorption test

Measuring no-load current absorption

The current absorbed by the vehicle system may be measured with a digital multimeter or with an amperometric clamp meter.

Use only amperometric clamp meters with mA scales.

The unit of measurement used for testing must be Amperes (A).

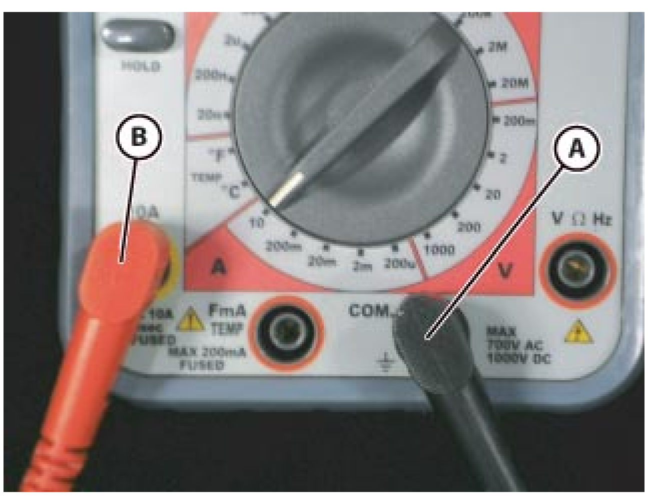
- Ensure that the doors and the luggage compartment lid are correctly closed.
- Open the engine compartment lid.

- Remove the engine compartment cosmetic shields (E3.14).

Remove the right hand cosmetic inspection shield.

PROCEDURE WITH DIGITAL MULTIMETER

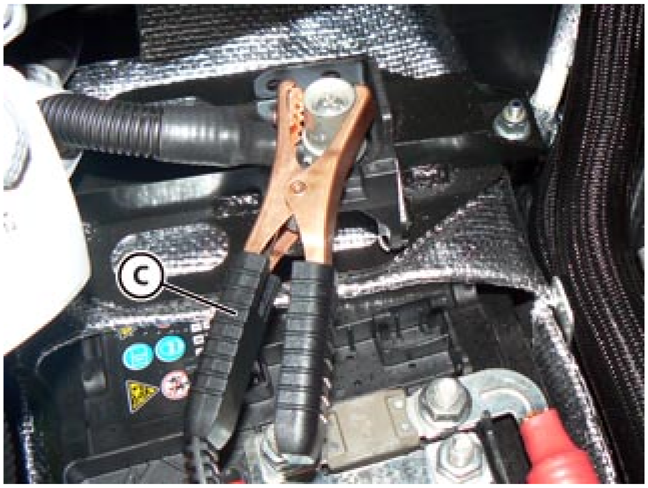
- Disconnect the battery (F2.01).



If possible, use cables with terminal clamps for the following procedures.

- Connect the wire **(A)** to the tester socket marked Ground.
- Connect the wire **(B)** (usually red) to the tester socket marked with the unit of measurement A (Amperes).

- Attach the lead clamp **(D)** onto the quick release terminal clamp connected to ground on the chassis.



Do not start the engine or turn the ignition key-on, as this may blow the instrument fuse.

- Attach the black lead clamp **(C)** (common ground) onto the false pole connected to the negative battery pole.

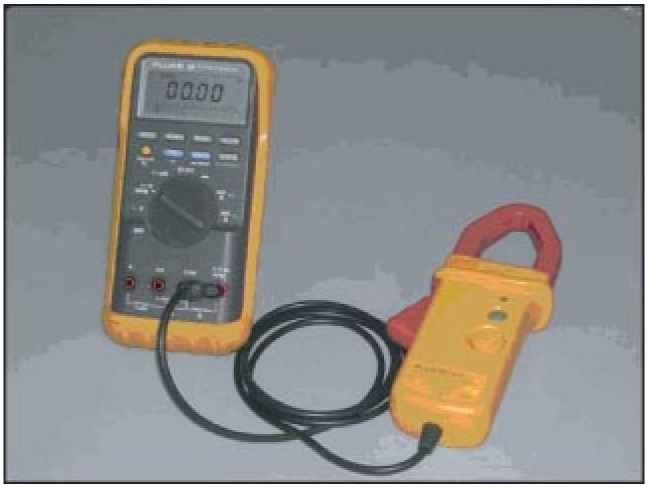
- To prevent the reading from being affected by devices that are still active, wait at least **15 minutes** for the network to enter **"Sleep Mode"**.
- Read the absorption values measured.
- The values read for zero-load current absorption must be close to **25 to 35 mA** and **MUST NOT** exceed **45 mA**.

- Remove the lead clamps.
- Connect the battery (F2.01).

PROCEDURE WITH AMPEROMETRIC CLAMP METER

- Disable the EPB electric parking brake Autohold function (D3.06).

- Close the driver side door ensuring that it closes correctly.



The instrument used must have a current scale precision in mA.

- Place the clamp meter unit around the ground cable leading from the negative battery pole.
- The direction of the current measured is indicated on the clamp.

As in this case the current originates from the battery and discharges on the chassis, the arrow indicates a direction leading away from the battery toward the earthing point on the chassis.

- To prevent the reading from being affected by devices that are still active, wait at least **6 minutes** for the network to enter **"Sleep Mode"**.
- The values read for zero-load current absorption must be close to **25 to 35 mA** and **MUST NOT** exceed **45 mA**.
- Refit the engine compartment cosmetic shields (E3.14).

Possible causes of excessive absorption

- If the value read in the vehicle is higher than in the specifications, the battery will tend to discharge rapidly.
- The cause of this is probably one or more devices failing to switch off.
- To identify the cause of excessive absorption, disconnect the networked ECUs one by one
- while observing the absorption value. This will drop when the active ECU is disconnected.
- Another possible cause for excessive current absorption may be a short circuit to ground at some point in the wiring harness.